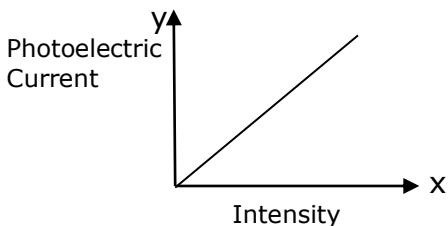
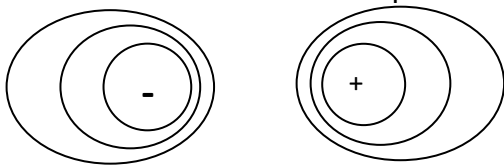


II PUC EXAMINATION MODEL QUESTION PAPER -3 (PU BOARD 2025-26)**SUBJECT : PHYSICS(33) SCHEME OF EVALUATION MAX MARKS :70**

Q.N	KEY ANSWERS	MARKS
PART- A		
I. Pick the correct option among the four given options for <u>ALL</u> of the following questions: <u>15x1=15</u>		
1.	The electrostatic force between a pair of unlike and like charges respectively are: (a) repulsive, attractive (b) attractive, repulsive (C)repulsive, repulsive (d) attractive, attractive ➤ Property of charges Answer: (b) attractive, repulsive	1
2.	The electric potential at the surface of a conducting charged spherical shell of radius 10 cm is 20V. The electric potential at the centre of the shell is: (a) zero (b) 10V (c) 20V (d) 30V ➤ The electric potential is constant throughout the volume of the shell. Therefore the electric potential at the centre is also 20V. Answer(c) 20V	1
3.	As the temperature increases, resistivity of: (a) metals increases and semiconductors decreases. (b) metals decreases and semiconductors increases. (c) both metals and semiconductors increases. (d) metals remains constant but semiconductors decreases. Temperature is directly proportional to the resistivity for metals. ➤ Temperature is inversely proportional to resistivity for semiconductors. Answer: (a) metals increases and semiconductors decreases.	1
4.	A charged particle is moving in a circular path in a uniform magnetic field. Then choose the wrong statement among the following: (a) Radius of the circular path varies directly with its velocity. (b) Radius of the circular path varies inversely with the magnetic field. (c) Frequency of revolution varies directly with the charge. (d) Frequency of revolution varies directly with the kinetic energy. $\text{➤ } r = \frac{mv}{Bq}$ $\text{➤ } f = \frac{Bqr}{m}$ Answer: (d) Frequency of revolution varies directly with the kinetic energy	1
5.	A magnetic dipole of magnetic moment $\vec{m}$ is placed in a uniform magnetic field $\vec{B}$ such that the angle between $\vec{m}$ and $\vec{B}$ is θ. If the magnetic dipole is in stable equilibrium position, then: (a) $\theta = 0^\circ$ (b) $\theta = 90^\circ$ (c) $\theta = 180^\circ$ (d) $\theta = 45^\circ$	1

	$\tau = mB \sin \theta$ for stable equilibrium $\theta = 0^0$ for unstable equilibrium $\theta = 180^0$ Answer: (a) $\theta = 0^0$									
6.	The dimensions of Inductance is: (a) $[ML^2T^{-2}A^{-2}]$ (b) $[M^2LT^{-2}A^{-2}]$ (c) $[ML^2T^{-2}A]$ (d) $[ML^2T^{-1}A^{-2}]$ $\triangleright$ The energy stored in the inductor is given by, $U = \frac{1}{2}LI^2$ $L = \frac{2U}{I^2}$ $[L] = \frac{[U]}{[I^2]} \quad (U=E)$ $[L] = \frac{[ML^2T^{-2}]}{[A^2]}$ $[L] = [ML^2T^{-2}A^{-2}]$ Answer: (a) $[ML^2T^{-2}A^{-2}]$	1								
7.	Different devices connected to an AC source are listed in the first column and different phase relations of electric current with the voltage are listed in the second column. Choose the correct match: <table><thead><tr><th>AC source connected to</th><th>Electric Current</th></tr></thead><tbody><tr><td>(i) resistor</td><td>(p) lags behind the voltage by $\frac{\pi}{2}$</td></tr><tr><td>(ii) inductor</td><td>(q) is in phase with voltage</td></tr><tr><td>(iii) capacitor</td><td>(r) leads the voltage by $\frac{\pi}{2}$</td></tr></tbody></table> (a) (i) $\rightarrow$ (p); (ii) $\rightarrow$ (q); (iii) $\rightarrow$ (r) (b) (i) $\rightarrow$ (r); (ii) $\rightarrow$ (p); (iii) $\rightarrow$ (q) (c) (i) $\rightarrow$ (q); (ii) $\rightarrow$ (p); (iii) $\rightarrow$ (r) (d) (i) $\rightarrow$ (q); (ii) $\rightarrow$ (r); (iii) $\rightarrow$ (p) Answer: (c) (i) $\rightarrow$ (q); (ii) $\rightarrow$ (p); (iii) $\rightarrow$ (r)	AC source connected to	Electric Current	(i) resistor	(p) lags behind the voltage by $\frac{\pi}{2}$	(ii) inductor	(q) is in phase with voltage	(iii) capacitor	(r) leads the voltage by $\frac{\pi}{2}$	1
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(ii) inductor	(q) is in phase with voltage									
(iii) capacitor	(r) leads the voltage by $\frac{\pi}{2}$									
8.	The expression for displacement current is given by(symbols have usual meanings): (a) $i_d = \epsilon_0 \frac{d\varphi_E}{dt}$ (b) $i_d = \mu_0 \frac{d\varphi_E}{dt}$ (c) $i_d = \mu_0 \epsilon_0 \frac{d\varphi_E}{dt}$ (d) $i_d = \frac{d\varphi_E}{dt}$ Answer: (a) $i_d = \epsilon_0 \frac{d\varphi_E}{dt}$	1								
9.	The objective lens of refracting telescope is a _____ and the magnifying power of the telescope for normal adjustment is _____ (a) convex lens of large aperture; $m = \frac{f_e}{f_o}$ (b) convex lens of small aperture; $m = \frac{f_e}{f_o}$ (c) convex lens of large aperture; $m = \frac{f_o}{f_e}$ (d) convex lens of small aperture; $m = \frac{f_o}{f_e}$	1								

	Answer: (c) convex lens of large aperture; $m = \frac{f_0}{f_e}$	
10.	Among the given two statements: Statement-I: Wavefront is the surface of constant phase. Statement II: When a plane wavefront is passed through a thin prism, it emerges out as a spherical wavefront. (a) Both the statements are correct (b) Both the statements are wrong (c) I is correct but II is wrong (d) I is wrong but II is correct ➤ When plane wave front enters the prism, it emerges as plane wave front only but changes its orientation Answer: (c) I is correct but II is wrong	1
11.	In photoelectric effect experiment, the graph of a physical quantity 'y' v/s a physical quantity 'x' is a straight line passing through origin. Then: (a) y is photoelectric current and x is collector plate potential. (b) y is photoelectric current and x is frequency of incident radiation. (c) y is stopping potential and x is frequency of incident radiation. (d) y is photoelectric current and x is intensity of incident radiation. ➤ From the laws of Photoelectric effect, intensity of incident radiation is directly proportional to the Photoelectric current  Answer: (d) y is photoelectric current and x is intensity of incident radiation.	1
12.	An α- particle, a proton, an electron and a neutron are moving with same velocity. Then the particle having longest de-Broglie wavelength is: (a) proton (b) electron (c) neutron (d) α- particle ➤ $\lambda = \frac{h}{mV} \Rightarrow \lambda \propto \frac{1}{m}$ ➤ For the particle whose mass is less, de-Broglie wavelength is longest $m_\alpha = 6.64 \times 10^{-27} \text{ Kg}$ $m_p = 1.67 \times 10^{-27} \text{ Kg}$ $m_e = 9.1 \times 10^{-31} \text{ Kg}$ $m_n = 1.67 \times 10^{-27} \text{ Kg}$ ➤ Since mass of electron is less, it has longest de-Broglie wavelength Answer: (b) electron	1
13.	Rutherford's nuclear model of atom fails to explain: (a) large angle scattering of α- particle.	

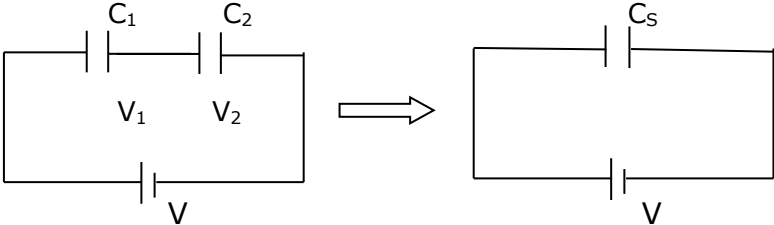
	(b) charge distribution in an atom. (c) energy of electrons revolving round the nucleus. (d) origin of line spectra of hydrogen atom. Answer: (d) origin of line spectra of hydrogen atom.	1
14.	Nuclear binding energy per nucleon is (A is the mass number of the nucleus): (a) more for heavy nuclei ($A > 170$). (b) more for lighter nuclei ($A < 30$). (c) less for both light and heavy nuclei. (d) independent of mass number. Answer: (c) less for both light and heavy nuclei.	1
15.	If n_e and n_h are the number of free electrons and holes respectively, then in the case of an intrinsic semiconductor: (a) $n_e = n_h$ (b) $n_e < n_h$ (c) $n_e > n_h$ (d) $n_e \gg n_h$ Answer: (a) $n_e = n_h$	1
II. Fill in the blanks by choosing appropriate answer given in the bracket for <u>ALL</u> the following questions: 5x1=5 (magnetic susceptibility, diffraction, electric field, magnetic field, equal to unity, total internal reflection)		
16.	An example for dimensionless quantity is <u>magnetic susceptibility</u> .	1
17.	A current carrying coil stores energy in the form of <u>magnetic field</u> .	1
18.	The power factor of a series LCR circuit at resonance is <u>equal to unity</u> .	1
19.	The principle of optical fibre is <u>total internal reflection</u> .	1
20.	The bending of light around the corners and entering into geometrical shadow region is called <u>diffraction</u> .	1
PART – B		
III. Answer any FIVE of the following questions: 5x2=10		
21.	Define electric flux through an area element. Mention its SI unit. Answer: - The total number of electric field lines passing through the given surface normally is called electric flux. - SI unit: Vm	1 1
22.	What is meant by equipotential surface? Draw equipotential surfaces for an electric dipole. Answer: The surface on which potential is same at all the points is called equipotential surface. The equipotential surfaces for an electric dipole: 	1 1
23.	Write any two limitations of Ohm's law. Answer: Ohm's law is applicable only for conductors.	

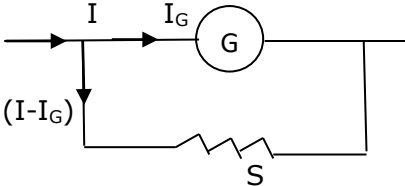
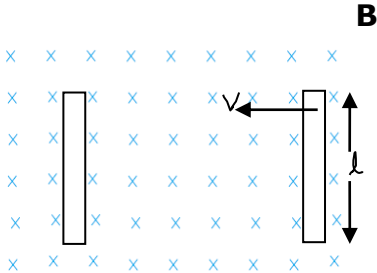
	• Ohm's law is applicable only when temperature is constant. • Ohm's law is applicable only when pressure is not applied on the conductor. • Ohm's law is not applicable for X-Ray and vacuum tubes. (Any Two)	2												
24.	Give the vector form of Biot-Savart's law and explain the terms. Answer: $\vec{dB} = \left(\frac{\mu_0}{4\pi}\right) \frac{I(\vec{dl} \times \vec{r})}{r^3}$ μ_0=permeability of free space I= current dl =length of current element r= distance between the current element and point 'P'	1 1												
25.	What is AC generator? What is its working principle? Answer: • AC generator is a device which is used to convert mechanical energy into electrical energy. • Principle:Electromagnetic Induction	1 1												
26.	List any two uses of infrared rays. Answer: • Used in remote control. • Used in solar cooker and heater. • Used in physiotherapy. (Any Two)	2												
27.	The radius of the innermost electron orbit of a hydrogen atom is 5.3×10^{-11} m. Determine the radius of the $n=2$ orbit. Answer: $r_1 = 5.3 \times 10^{-11}$m; For $n=2$, $r_2 = ?$ For H-atom, $r_n = \frac{\epsilon_0 n^2 h^2}{\pi m Z e^2}$ $\therefore r_n = n^2 \times r_1$ For $n=2$ $\therefore r_2 = 2^2 \times 5.3 \times 10^{-11}$ $r_2 = 4 \times 5.3 \times 10^{-11}$ $r_2 = 21.2 \times 10^{-11}\text{m}$ (correct answer with unit is necessary)	1 1												
28.	Mention any two differences between half wave and full wave rectifiers. Answer: <table><tr><th>S.N</th><th>Half wave rectifier</th><th>S.N</th><th>Full wave rectifier</th></tr><tr><td>1</td><td>It converts only positive half cycle of input AC into DC.</td><td>1</td><td>It converts both positive half cycle and negative half cycle of input AC into DC.</td></tr><tr><td>2</td><td>It uses only one diode.</td><td>2</td><td>It uses two diodes.</td></tr></table>	S.N	Half wave rectifier	S.N	Full wave rectifier	1	It converts only positive half cycle of input AC into DC.	1	It converts both positive half cycle and negative half cycle of input AC into DC.	2	It uses only one diode.	2	It uses two diodes.	1 1
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PART-C

IV. Answer any FIVE of the following questions:

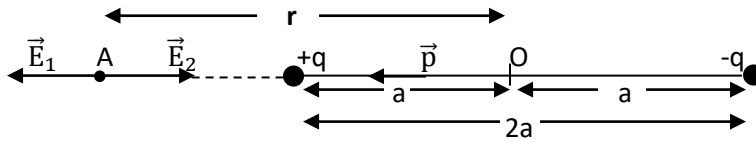
5x3=15

29.	Define linear charge density. Mention the expression for electric field due to an infinite line of charge and explain the terms. Answer:The charge per unit length of the conductor is called linear charge density. $E = \frac{\lambda}{2\pi\epsilon_0 r}$ λ = linear charge density ϵ_0 = permittivity of free space r = distance between the straight conductor and point 'P'	1 1 1
30.	Obtain the expression for equivalent capacitance of two capacitors connected in series. Answer:  When capacitors are connected in series, - Charge on each capacitor is same. - Total potential of series combination is equal to sum of potential across individual capacitor That is, $V = V_1 + V_2 \dots \dots (1)$ We know that, $\Rightarrow \left. \begin{aligned} Q &= CV \\ V &= \frac{Q}{C} \\ V_1 &= \frac{Q}{C_1} \\ V_2 &= \frac{Q}{C_2} \end{aligned} \right\} \longrightarrow (2)$ (2) in (1) we get, $\frac{Q}{C} = \frac{Q}{C_1} + \frac{Q}{C_2}$ $\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$ Here $C = C_s$ = Equivalent capacitance of series combination Therefore, $\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2}$ OR	1 1 1

	$C_s = \frac{C_1 C_2}{C_1 + C_2}$ C_1 and C_2 are the capacitance of individual capacitors V = total potential Q = charge on each capacitor																
31.	Explain the conversion of galvanometer into an ammeter using relevant circuit diagram and expression. Answer: Galvanometer can be converted into ammeter by connecting very small resistance in parallel with it  Potential across S = Potential across G $(I - I_G)S = I_G G$ $S = \frac{I_G G}{I - I_G}$ I_G = current through the galvanometer I = total current G = resistance of the galvanometer	1 1 1															
32.	Write any three differences between diamagnetic and paramagnetic materials. Answer: <table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th>SN</th><th>Diamagnetic materials</th><th>Paramagnetic materials</th></tr> </thead> <tbody> <tr> <td>1</td><td>They are weakly repelled by the bar magnet.</td><td>They are weakly attracted by the bar magnet</td></tr> <tr> <td>2</td><td>$\mu_r < 1$</td><td>$\mu_r > 1$</td></tr> <tr> <td>3</td><td>Susceptibility is low and negative</td><td>Susceptibility is low and positive</td></tr> <tr> <td>4</td><td>EX:- Gold, Silver, Diamond</td><td>EX:- Sodium, Potassium</td></tr> </tbody> </table>	SN	Diamagnetic materials	Paramagnetic materials	1	They are weakly repelled by the bar magnet.	They are weakly attracted by the bar magnet	2	$\mu_r < 1$	$\mu_r > 1$	3	Susceptibility is low and negative	Susceptibility is low and positive	4	EX:- Gold, Silver, Diamond	EX:- Sodium, Potassium	3
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3	Susceptibility is low and negative	Susceptibility is low and positive															
4	EX:- Gold, Silver, Diamond	EX:- Sodium, Potassium															
33.	Derive the expression for the motional emf induced in a conducting rod moving perpendicular to a uniform Magnetic field. Answer: EMF induced across the ends of the conductor when conductor is made to move in perpendicular magnetic field is called motional emf. 	1															

	$(2) \Rightarrow 2i = \frac{PM}{PF}$ -----(4) $(3) \div (4)$ $\frac{i}{2i} = \frac{\left(\frac{PM}{PC}\right)}{\left(\frac{PM}{PF}\right)}$ $\frac{1}{2} = \frac{PF}{PC}$ But $PF = f$, $PC = R$ $\therefore \frac{1}{2} = \frac{f}{R}$ $\therefore f = \frac{R}{2}$ F=focal length R=radius of curvature (No arrows, deduct one mark)	1 1
35.	The work function of cesium is 2.114 eV. Calculate the threshold frequency for cesium. Answer: $W = 2.14 \text{ eV}$; $\gamma_0 = ?$; $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$ $W = 2.14 \times 1.6 \times 10^{-19} \text{ J}$ $W = h\gamma_0$ $2.14 \times 1.6 \times 10^{-19} = 6.625 \times 10^{-34} \times \gamma_0$ $\gamma_0 = \frac{3.424 \times 10^{-19}}{6.625 \times 10^{-34}}$ $\gamma_0 = 0.51683 \times 10^{15} \text{ Hz}$ $\gamma_0 = 5.1683 \times 10^{14} \text{ Hz}$	1 1 1
36.	Mention three features of nuclear force. Answer: • Nuclear force is the strongest force in nature. • Nuclear force is attractive in nature. • Nuclear force is saturated force. • Nuclear force is charge independent. • Nuclear force is spin dependent. • Nuclear force is exchange force. • Nuclear force is non-central force. (Any Three)	3
PART – D		
V. Answer any THREE of the following questions:		3x5=15
37.	Derive an expression for the electric field at a point on the axis of an electric dipole.	

Answer:



1

Electric field at a point 'A' due to charge +q is given by,

$$E_1 = \frac{1}{4\pi\epsilon_0} \frac{q}{(r-a)^2}$$

Electric field at a point 'A' due to charge -q is given by,

$$E_2 = \frac{1}{4\pi\epsilon_0} \frac{q}{(r+a)^2}$$

Resultant electric field at point 'A' is given by,

1

$$E = E_1 + E_2$$

Since electric fields E_1 and E_2 are opposite to each other

$$E = E_1 - E_2$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{(r-a)^2} - \frac{1}{4\pi\epsilon_0} \frac{q}{(r+a)^2}$$

$$E = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{(r-a)^2} - \frac{1}{(r+a)^2} \right]$$

$$E = \frac{q}{4\pi\epsilon_0} \left[\frac{(r+a)^2 + (r-a)^2}{(r-a)^2(r+a)^2} \right]$$

1

$$E = \frac{q}{4\pi\epsilon_0} \left[\frac{r^2 + a^2 + 2ar - r^2 - a^2 + 2ar}{(r-a)^2(r+a)^2} \right]$$

WKT $r^2 - a^2 = (r+a)(r-a)$ therefore $(r+a)^2(r-a)^2 = (r^2 - a^2)^2$

$$E = \frac{q}{4\pi\epsilon_0} \frac{4ar}{(r^2 - a^2)^2}$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{(2aq)2r}{(r^2 - a^2)^2}$$

1

But, $2aq = p$

Therefore,

$$E = \frac{1}{4\pi\epsilon_0} \frac{p(2r)}{(r^2 - a^2)^2}$$

If r^2 is >>>>>> a^2 , a^2 can be neglected.

$$E = \frac{1}{4\pi\epsilon_0} \frac{2p}{r^3}$$

1

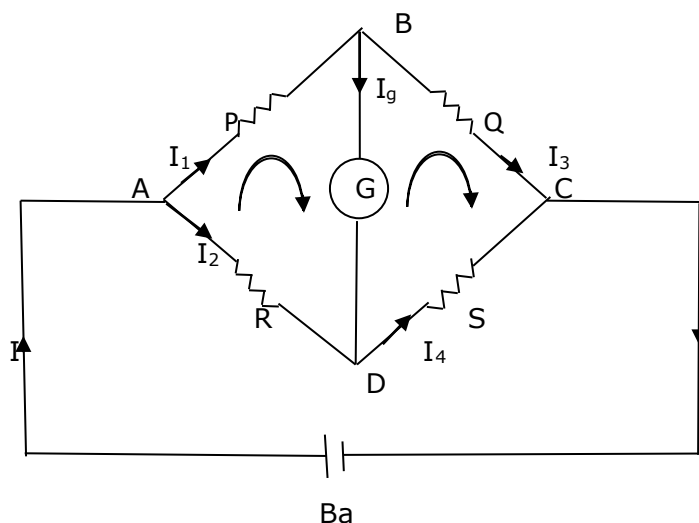
p =electric dipole moment

ϵ_0 =permittivity of free space

r =distance between A and O

38. Using Kirchhoff's laws, arrive at the balancing condition of Wheatstone network.

Answer:



1

Applying Kirchhoff's Current Rule at node 'B'

$$I_1 - I_g - I_3 = 0$$

$$I_1 = I_g + I_3 \text{ --- (1)}$$

Applying Kirchhoff's Current Rule at node 'D'

$$I_2 + I_g - I_4 = 0$$

$$I_4 = I_g + I_2 \text{ --- (2)}$$

Applying Kirchhoff's Voltage Rule for loop ABDA

$$I_1 P - I_2 R + I_g G = 0$$

$$I_2 R = I_1 P + I_g G \text{ --- (3)}$$

Applying Kirchhoff's Voltage Rule for loop BCDB

$$I_3 Q - I_4 S - I_g G = 0$$

$$I_3 Q = I_4 S + I_g G \text{ --- (4)}$$

Wheatstone network is balanced when $I_g = 0$

$$\therefore (3) \Rightarrow I_1 P = I_2 R \text{ --- (5)}$$

$$(4) \Rightarrow I_3 Q = I_4 S \text{ --- (6)}$$

$$(1) \Rightarrow I_1 = I_3$$

$$(2) \Rightarrow I_2 = I_4$$

Dividing (5) by (6)

$$\frac{I_1 P}{I_3 Q} = \frac{I_2 R}{I_4 S}$$

$$\boxed{\therefore \frac{P}{Q} = \frac{R}{S}}$$

1

1

1

1

(No diagram No mark and wrong diagram OR wrong direction of current deduct one mark)

**39. i) State and explain Ampere's circuital law.
ii) Derive the expression for the magnetic field at a point due to an infinitely long straight current carrying wire using Ampere's circuital law.**

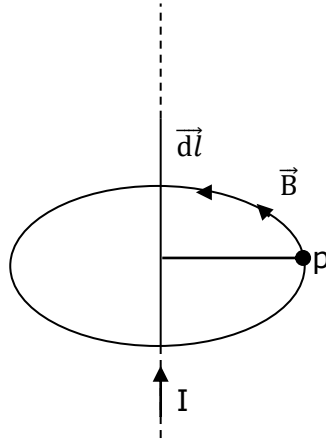
Answer:

i) "Line integral of magnetic field around closed path is equal to μ_0 times the total current flowing through the closed path"

$$\oint_l \vec{B} \cdot d\vec{l} = \mu_0 I$$

I = current, μ_0 = permeability of free space

ii)



I=current

r=distance between conductor and 'p'

From Ampere's circuital law,

$$\oint_l \vec{B} \cdot d\vec{l} = \mu_0 I$$

$$\oint_l B \cdot dl \cos \theta = \mu_0 I$$

But $\theta = 0^\circ$

$$\therefore \oint_l B \cdot dl = \mu_0 I$$

$$B \oint_l dl = \mu_0 I$$

But $\oint dl = \text{circumference} = 2\pi r$

$$\therefore B(2\pi r) = \mu_0 I$$

$$B = \frac{\mu_0 I}{2\pi r}$$

2

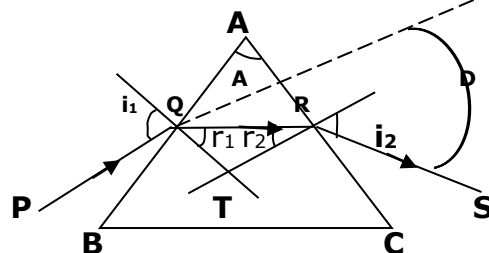
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1

40. Derive the expression for refractive index of the material of a prism in terms of angle of the prism and angle of minimum deviation.

Answer:



From Snell's law

$$n = \frac{\sin i}{\sin r} \quad \text{---(1)}$$

Deviation at surface AB is

$$d_1 = i_1 - r_1$$

Deviation at surface AC is

$$d_2 = i_2 - r_2$$

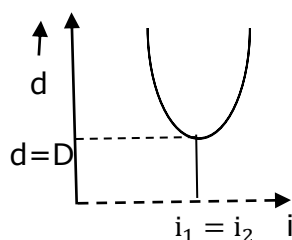
Total deviation is

$$d = d_1 + d_2$$

$$d = (i_1 - r_1) + (i_2 - r_2)$$

$$d = (i_1 + i_2) - (r_1 + r_2) \quad \text{-----(2)}$$

From i-d graph,



At minimum deviation

$$\left. \begin{array}{l} i_1 = i_2 = i \\ d = D, r_1 = r_2 = r \end{array} \right\} \rightarrow (3)$$

From triangle QTR

$$r_1 + r_2 + \angle QTR = 180^\circ \quad \text{---(4)}$$

From Quadrilateral

$$A + \angle QTR = 180^\circ \quad \text{----(5)}$$

From (4) and (5)

$$r_1 + r_2 = A \quad \text{--- (6)}$$

But $r_1 = r_2 = r$

$$\therefore 2r = A$$

$$\boxed{r = \frac{A}{2}} \quad \text{---(7)}$$

(3) & (6) in (2)

$$\therefore D = 2i - A$$

$$i = \frac{A+D}{2} \quad \text{--- (8)}$$

(7) & (8) in (1)

1

1

1

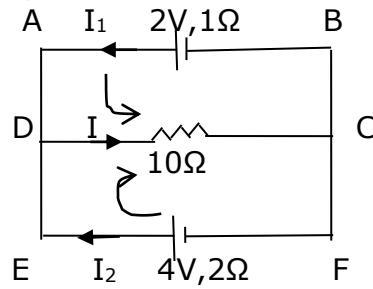
1

	$\therefore n = \frac{\sin\left(\frac{A+D}{2}\right)}{\sin\left(\frac{A}{2}\right)}$ (No arrows, deduct one mark)	1																
41	i)Explain the formation of energy bands in solids. ii)On the baaasis of energy bands,distinguish between a conductor, a semiconductor and an insulator. Answer: i) For an isolated atom, electron posess discrete energy level. When number of atoms comes together, energy level of an electrons overlaps and posess range of energy values called energy bands. ii) <table><tr><th>S.N</th><th>CONDUCTORS</th><th>INSULATORS</th><th>SEMICONDUCTORS</th></tr><tr><td>1</td><td>Conductors are the materials in which there is an overlapping between Conduction band and Valence band.</td><td>Insulators are the materials in which there is a large energy gap of order $E_g > 3\text{eV}$ between Conduction band and Valence band.</td><td>Semiconductors are the materials in which there is a small energy gap of order $E_g < 3\text{ eV}$ between Conduction band and Valence band.</td></tr><tr><td>2</td><td> CB VB </td><td> CB $E_g>3\text{eV}$ VB </td><td> CB $E_g<3\text{ eV}$ VB </td></tr><tr><td>3</td><td>Example: All metals</td><td>Ex: Wood, Glass etc</td><td>Ex:Silicon, Germanium</td></tr></table>	S.N	CONDUCTORS	INSULATORS	SEMICONDUCTORS	1	Conductors are the materials in which there is an overlapping between Conduction band and Valence band.	Insulators are the materials in which there is a large energy gap of order $E_g > 3\text{eV}$ between Conduction band and Valence band.	Semiconductors are the materials in which there is a small energy gap of order $E_g < 3\text{ eV}$ between Conduction band and Valence band.	2	CB VB	CB $E_g>3\text{eV}$ VB	CB $E_g<3\text{ eV}$ VB	3	Example: All metals	Ex: Wood, Glass etc	Ex:Silicon, Germanium	2 3
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VI. Answer any TWO of the following questions:2 × 5 =10 42. Three point charges $3nC$, $-2nC$ and $4nC$ are placed at the vertices A,B and C of an equilateral triangle ABC of sides 0.2 m. Calculate the potential energy of the system. Also calculate the amount of work required to place the same charges at the vertices of an equilateral triangle of side 0.1 m.																		

	(Take: $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2 \text{ C}^{-2}$) Answer: U=?, W=? $U_i = \left(\frac{1}{4\pi\epsilon_0} \right) \left[\frac{q_1 q_2}{r_{12}} + \frac{q_2 q_3}{r_{23}} + \frac{q_3 q_1}{r_{31}} \right]$ Here $r_{12}=r_{23}=r_{31}=0.2 \text{ m}$ (Equilateral triangle) $U_i = \frac{9 \times 10^9}{0.2} [(3 \times -2) + (-2 \times 4) + (4 \times 3)] \times 10^{-18}$ $U_i = \frac{9 \times 10^{-9}}{0.2} [-6 - 8 + 12]$ $U_i = -9 \times 10^{-8} \text{ J}$ $U_i = -90 \times 10^{-9} \text{ J}$ Now $U_f = \left(\frac{1}{4\pi\epsilon_0} \right) \left[\frac{q_1 q_2}{r_{12}} + \frac{q_2 q_3}{r_{23}} + \frac{q_3 q_1}{r_{31}} \right]$ $U_f = \frac{9 \times 10^9}{0.1} [(3 \times -2) + (-2 \times 4) + (4 \times 3)] \times 10^{-18}$ $U_f = -180 \times 10^{-9} \text{ J}$ From Work-Energy theorem, $W = \Delta U$ $W = U_f - U_i$ $W = (-180 \times 10^{-9}) - (-90 \times 10^{-9})$ $W = -90 \times 10^{-9} \text{ J}$ Since work is to be done to arrange the charges, energy will be spent. $\therefore$ work done should be negative. $W = -(-90 \times 10^{-9} \text{ J})$ $W = 90 \times 10^{-9} \text{ J}$	1 2 1 1
43.	Calculate the power dissipated in the 10Ω resistor in the given circuit.	

Answer:

From the figure,



$$P_{10} = I^2 R$$

1

From KCR, at node D, $I = I_1 + I_2$

From KVL for loop ABCDA,

$$\sum \varepsilon = \sum IR$$

$$2 = 10I + I_1 \text{ -----(1)}$$

1

For loop DCFED,

$$4 = 10I + 2I_2 \text{ -----(2)}$$

Equation (2)-(1) we get

$$2 = 2I_2 - I_1 \text{ -----(3)}$$

But $I_2 = I - I_1$

$$2 = 2(I - I_1) - I_1$$

$$\therefore 2 = 2I - 2I_1 - I_1$$

1

$$2 = 2I - 3I_1 \text{ -----(4)}$$

Equation (4) + (1)

$$2 = 2I - 3I_1$$

$$\frac{(2 = 10I + I_1) \times 3}{8 = 32I}$$

$$\therefore I = \frac{1}{4}$$

1

$$I = 0.25A$$

Now

$$P_{10} = I^2 R$$

$$P_{10} = (0.25)^2 \times 10$$

$$P_{10} = 0.625 W$$

1**ALTERNATE METHOD**

$$\varepsilon_1 = 2V, \varepsilon_2 = 4V$$

$$r_1 = 1\Omega, r_2 = 2\Omega$$

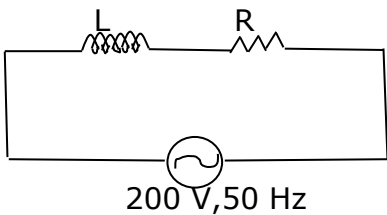
$$R = 10\Omega, P_{10} = ?$$

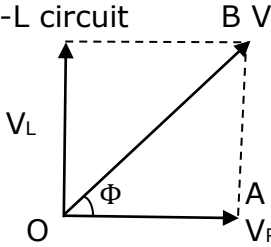
$$P_{10} = I^2 R$$

1

$$I = ?$$

$$I_{eq} = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}$$

	$\varepsilon_{eq} = \frac{(2 \times 2) + (4 \times 1)}{2 + 1}$ $\boxed{\varepsilon_{eq} = 2.667 \text{ V}}$ $\boxed{r_{eq} = \frac{r_1 r_2}{r_1 + r_2}}$ $r_{eq} = \frac{1 \times 2}{1 + 2}$ $\boxed{r_{eq} = 0.667 \Omega}$ WKT, $V = \varepsilon_{eq} - I r_{eq}$ But $V = IR$ $\therefore IR = \varepsilon_{eq} - I r_{eq}$ $I \times 10 = 2.667 - I \times 0.667$ $10I + 0.667I = 2.667$ $10.667I = 2.667$ $\boxed{I = 0.25 \text{ A}}$ $P_{10} = I^2 R$ $P_{10} = (0.25)^2 \times 10$ $\boxed{P_{10} = 0.625 \text{ W}}$	1 1 1 1
44.	An inductor and a resistor are connected in series with 200 V, 50 Hz ac source. The current in the circuit is 2A and voltage leads the current by $\frac{\pi}{3}$. Calculate the inductance of the inductor. Answer:  $I = 2 \text{ A}, \varphi = \frac{\pi}{3}, L = ?$ $\boxed{X_L = L\omega}$ From Ohm's law, $V = IZ$ $Z = \frac{V}{I}$	1

	$Z = \frac{200}{2}$ $Z = 100\Omega$ Phasor diagram for R-L circuit  From phasor diagram, $\sin \phi = \frac{V_L}{V}$ $\sin \phi = \frac{IX_L}{IZ}$ $\sin \phi = \frac{X_L}{Z}$ $\therefore X_L = Z \times \sin \phi$ $X_L = 100 \times \sin \frac{\pi}{3}$ $X_L = 100 \times \sin 60^\circ$ $X_L = 100 \times \frac{\sqrt{3}}{2}$ $X_L = 86.6\Omega$ Now $X_L = L \times 2\pi f$ $L = \frac{X_L}{2\pi f}$ $L = \frac{86.6}{2 \times 3.14 \times 50}$ $L = 0.275 \text{ H}$	1 1 1 1
45.	In Young's double slit experiment, the slits are separated by a distance of 0.5 mm and the screen is at a distance of 1.5 m from the slits. If the distance of 9th bright fringe from the central maximum is 1.3 cm, Calculate the wavelength of light used. Also calculate the distance of 5th dark fringe from the central maximum. Answer: $d = 0.5 \text{ mm}$, $D = 1.5 \text{ m}$, $X_9 = 1.3 \text{ cm} \Rightarrow$ Bright fringe $\lambda = ?$, $x_5 = ?$ For dark fringe Distance of n^{th} bright fringe is: $x_n = \frac{n\lambda D}{d}$ ($n=1,2,3,\dots$)	1

	$\lambda = \frac{x_n \times d}{nD}$ For $n=9$, $\lambda = \frac{1.3 \times 10^{-2} \times 0.5 \times 10^{-3}}{9 \times 1.5}$ $\lambda = 481.48 \times 10^{-9} \text{m}$	2
	Distance of n^{th} dark fringe is $x_n = \frac{(2n-1)\lambda D}{2d} \quad (n=1,2,3, \dots)$	1
	$x_5 = \frac{(2 \times 5 - 1) \times 481.48 \times 10^{-9} \times 1.5}{2 \times 0.5 \times 10^{-3}}$ $x_5 = 6.499 \times 10^{-3} \text{m}$	1